

SENSOR-BASED SIMULATION

LED On/Off Using a Sensor

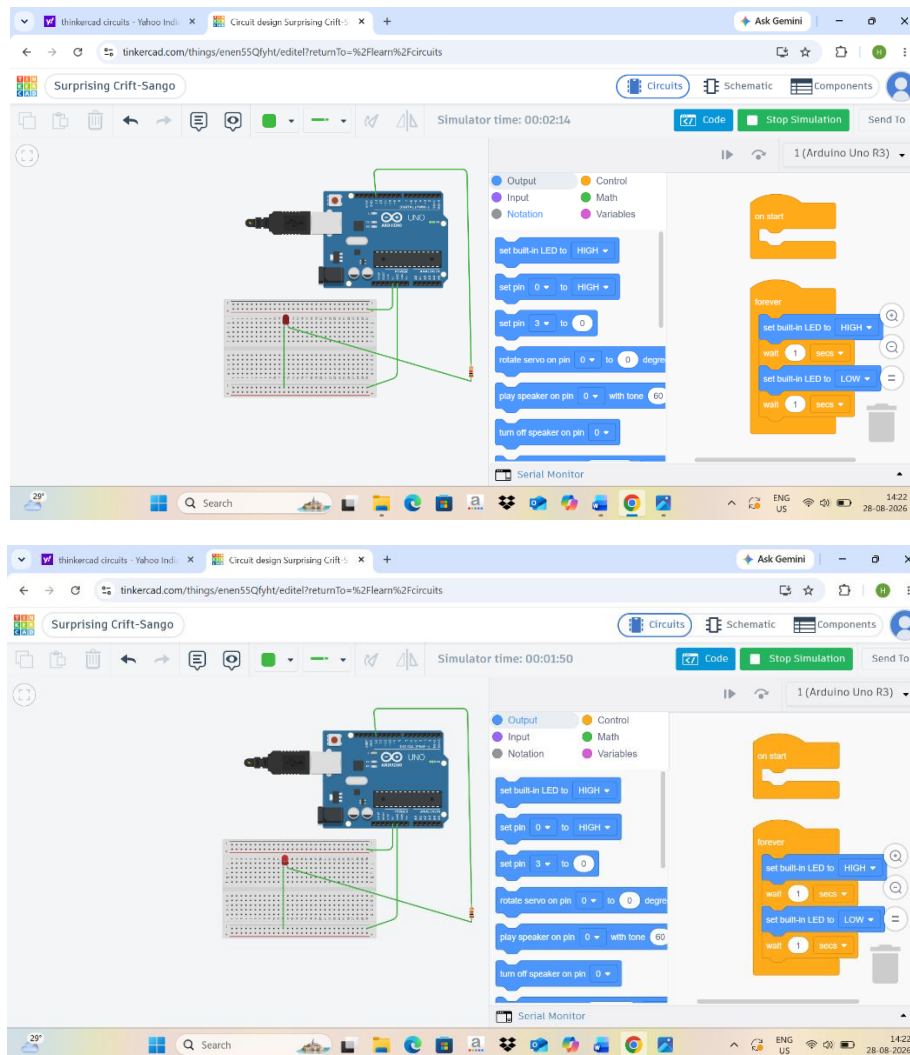


Fig : Simulation of LED

The block code shown corresponds to turning the LED ON and OFF every 1 second.

1.Arduino code

```
int LED = 13;

void setup()
{
  pinMode(LED, OUTPUT);
}
```

```
void loop()
{
  digitalWrite(LED, HIGH);
  delay(1000);
  digitalWrite(LED, LOW);
  delay(1000);
}
```

2.Explanation

1. int LED = 13;

We assign Arduino digital pin 13 to the LED.

2. void setup()

- This part runs only once when the Arduino starts.
pinMode(LED, OUTPUT);
- It configures pin 13 as an output pin.

3. void loop()

- This part runs continuously.
digitalWrite(LED, HIGH);
- Turns the LED ON.
delay(1000);
- Waits for 1000 milliseconds = 1 second.
digitalWrite(LED, LOW);
- Turns the LED OFF.
delay(1000);
- Waits another 1 second.
- Then loop() starts again

3.Working

LED ON → 1 second → LED OFF → 1 second → repeat continuously

Components used

1. Arduino UNO
2. LED
3. Resistor (typically 220 Ω)
4. Breadboard
5. Jumper wires